

Karnataka University Dharwad

B.Sc. -I Semester (RC-NEP) DSC Mathematics w.e.f. 2026-27

Course Code: CC1MAT1T1

Course Title: Algebra-I and Calculus-I (DSC-1)

(Karnatak University Mathematics Teacher's Association, Dharwad)

Question Bank

Unit-I: Matrices

Two Marks Questions:

- 1) State any two elementary row transformations or operations.
- 2) State any two elementary column transformations or operations.
- 3) Define rank of a matrix.
- 4) Define Eigen values of a square matrix A.
- 5) Define characteristic vector of a square matrix A.
- 6) State Cayley Hamilton Theorem.
- 7) If $A = \begin{bmatrix} 6 & x & -1 \\ 2 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix}$ and $\rho(A) = 2$, find the value of x .
- 8) Find the rank of the matrix $\begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 6 \\ 3 & 1 & 4 \end{bmatrix}$
- 9) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 7 & 8 & 10 \end{bmatrix}$
- 10) Verify whether the following system of equation has trivial or non trivial solution
 - 1) $x - y + 2z = 0$; $4x + y + 5z = 0$; $3x + 2y + z = 0$
 - 2) $-x + y + z = 0$; $3x + y - z = 0$; $2x + 2y = 0$
 - 3) $x - 3y + 2z = 0$; $7x - 21y + 14z = 0$; $-3x + 9y - 6z = 0$
- 11) Write the criterion for consistency of non-homogeneous system of linear equation
- 12) Find the Eigen values of the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$

Unit-II Limits and Continuity

Two marks questions:

- 1) Define a continuity of a function at a point using ϵ, δ
- 2) Discuss the continuity of $f(x) = \begin{cases} \frac{2}{5-x}, & \text{for } x < 3 \\ 5-x, & \text{for } x \geq 3 \end{cases}$ at $x=3$.
- 3) i) If $f(x) = \begin{cases} 5x-4, & \text{when } x \leq 1 \\ 4x^3-3x, & \text{when } 1 < x < 2 \end{cases}$ then show that $f(x)$ is continuous at $x = 1$
(ii) If $f(x) = \begin{cases} 3x-2, & \text{for } x \leq 1 \\ 2x-1, & \text{for } x > 1 \end{cases}$ then show that $f(x)$ is continuous at $x=1$.
(iii) If $f(x) = \begin{cases} 4x+3, & \text{for } x < 4 \\ 3x+7, & \text{for } x \geq 4 \end{cases}$ then show that $f(x)$ is continuous at $x = 4$.
- 4) Show that every derivable function at a point is continuous at that point.

- 5) (i) If $f(x) = \begin{cases} \lambda x^2, & \text{for } x \leq 2 \\ 3, & \text{for } x > 2 \end{cases}$ is continuous at $x = 2$, find the value of λ .
- (ii) If $f(x) = \begin{cases} 3x - 2, & \text{for } x \leq 1 \\ k, & \text{for } x > 1 \end{cases}$ is continuous at $x = 1$, find the value of k .
- (iii) If $f(x) = \begin{cases} \frac{\sin 3x}{x}, & x \neq 0 \\ K, & x = 0 \end{cases}$ is continuity at $x = 0$, then find K .
- 6) If $f(x)$ is function defined as $f(x) = \begin{cases} x - 2, & \text{if } x \leq 1 \\ x + 2, & \text{if } x > 1 \end{cases}$ then prove that $f(x)$ is discontinuous at $x = 1$.
- 7) State intermediate value theorem.
- 8) Evaluate
- i) $\lim_{x \rightarrow 0} \frac{(x - \sin x)}{x^3}$ ii) $\lim_{x \rightarrow \frac{\pi}{2}} (\sec x - \tan x)$
- iii) $\lim_{x \rightarrow 0} x \log x$ iv) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x}$
- v) $\lim_{x \rightarrow \infty} \frac{\log x}{x}$ vi) $\lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$
- vii) $\lim_{x \rightarrow 0} \frac{(1 - \cos x)}{x^2}$ viii) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\tan x}$

Unit-III: Polar Coordinates

Two Marks Questions:

- Find the angle between radius vector and tangent to the curve $r = a \sin \theta$.
 - Find the angle between radius vector and tangent to the curve $r = a(1 - \cos \theta)$.
- Find ϕ for the curve $r^2 = a^2 \cos 2\theta$ at $\theta = \pi/6$
 - Find ϕ for the curve $r = a(1 + \cos \theta)$ at $\theta = \pi/6$
- If $r = 4 \cos \theta$ then show that $\phi = \frac{\pi}{2} + \theta$.
- With usual notations prove that $p = r \sin \phi$
- Show that in the equiangular spiral $r = ae^{\theta \cot \alpha}$ the tangent is inclined at a constant angle to the radius vector.
- For the curve $r = a\theta$, show that $p = \frac{r^2}{\sqrt{r^2 + a^2}}$
- Define curvature of a curve at any point on the curve.
- Define radius of curvature of a curve at any point on the curve.
- Find the radius of curvature of the curve $y = c \log \sec x$ at any point (x, y) .
 - Find the radius of curvature of the curve $y = e^x$ at the point where it crosses y-axis
 - Find the radius of curvature for the curve $y = x^3$ at $x=1$
- Find the points of inflexion of the curve $y = xe^x$
Or Find the points of inflexion of the curve $x = \log\left(\frac{y}{x}\right)$
- Show that the curve $y = x^2$ has no points of inflexion
- Show that $y = e^x$ is concave upwards over $(-\infty, \infty)$.
Or Show that $y = e^x$ is everywhere concave upwards.
 - Show that $y = \log x$ is convex upwards over $(1, \infty)$.
Or Show that $y = \log x$ is convex upwards everywhere.

Unit-IV Successive differentiation and Mean Value Theorems

Two marks questions:

- 1) Find the n^{th} derivative of i) $\sin^2 x$ ii) $\cos^2 x$ iii) $\sin 3x \cos 2x$
- 2) Find the n^{th} derivative of i) $\cos(ax + b)$ ii) $\sin(ax + b)$ iii) e^{ax+b}
- 3) Show that $D^n(x^n) = n!$ where $D = \frac{d}{dx}$
- 4) State Rolle's theorem.
- 5) State Lagrange's mean value theorem.
- 6) State Cauchy's mean value theorem.
- 7) State Taylor's theorem.
- 8) (i) Expand e^x in powers of x by Maclaurin's expansion.
(ii) Expand $\cos x$ in powers of x by Maclaurin's expansion.
(iii) Expand $\sin x$ in powers of x by Maclaurin's expansion.

Unit-I: Matrices

Five marks questions:

- 1) Prove that the interchange of any two rows of a matrix does not alter the rank of a matrix.
- 2) Prove that the rank of a matrix is unaltered by multiplying the elements of a row by a non-zero scalar.
- 3) Prove that transpose of a matrix does not alter the rank.
Or Prove that $\rho(A) = \rho(A')$ where A is a matrix.
- 4) Find the rank of the following matrices by reducing to Echelon form.

i) $\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 5 & 2 \end{bmatrix}$ ii) $\begin{bmatrix} 1 & 3 & 4 & 5 \\ 1 & 2 & 6 & 7 \\ 1 & 5 & 0 & 1 \end{bmatrix}$ iii) $\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$

iv) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 1 & 4 & 7 & 10 \end{bmatrix}$ v) $\begin{bmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{bmatrix}$

- 5) Find the rank of the following matrices by reducing to normal form.

i) $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \end{bmatrix}$ ii) $\begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \end{bmatrix}$ iii) $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & 3 & 2 & 1 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \end{bmatrix}$

iv) $\begin{bmatrix} 3 & -1 & -1 & 2 \\ 2 & 2 & 1 & -1 \\ 1 & -3 & 2 & 1 \\ 1 & -3 & 6 & -1 \end{bmatrix}$ v) $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 3 & 4 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$ (vi) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 1 & 4 & 7 & 10 \end{bmatrix}$

(vii) $\begin{bmatrix} 1 & 1 & 3 & 1 \\ 5 & 1 & 5 & 3 \\ 2 & 0 & 1 & 1 \\ 0 & 2 & 5 & 1 \end{bmatrix}$ (viii) $\begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{bmatrix}$

6) Find the inverse of the following matrices using Cayley-Hamilton theorem.

(i) $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & -2 \\ 5 & -4 \end{bmatrix}$ iv) $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

7) Solve the following system of equations

1) $x + y + z = 2$; $2x - y + 3z = 8$; $3x + 2y - z = 3$

2) $x + y + z = 1$; $x + 2y + 4z = 2$; $x + 4y + 10z = 4$

3) $x + 3y - 2z = 0$; $2x - y + 4z = 0$; $x - 11y + 14z = 0$

4) $-x + y + z = 0$; $3x + y - z = 0$; $2x + 2y = 0$

5) $x + 3y + 6z = 2$; $3x - y + 4z = 9$; $x - 4y + 2z = 7$

8) Find the Eigen value and Eigen Vectors of

i) $\begin{bmatrix} 3 & 4 \\ 3 & 2 \end{bmatrix}$ ii) $\begin{bmatrix} 4 & 3 \\ 2 & 9 \end{bmatrix}$ iii) $\begin{bmatrix} 5 & -1 \\ 4 & 9 \end{bmatrix}$ iv) $\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

9) (i) Investigate for what values of λ and μ the equations:

$$x + y + z = 6; x + 2y + 3z = 10; x + 2y + \lambda z = \mu$$

have i) No solution ii) Unique solution and iii) Infinitely many solution.

(ii) For what values of λ and μ the equations:

$$x + 2y + 3z = 5; x + 3y - z = 4; x + 4y + \lambda z = \mu,$$

have i) No solution ii) Unique solution and iii) Infinitely many solution.

(iii) Investigate for what values of λ and μ the equations:

$$x + y + z = 6; x + 2y + 3z = 10; x + 2y + \lambda z = \mu$$

have i) No solution ii) Unique solution iii) Infinitely many solution.

(iv) Investigate for what values of λ and μ the equations:

$$x - 2y + 3z = 1; 3x + y - z = 4; 2x - 2y + \lambda z = \mu$$

have i) No solution ii) Unique solution iii) Infinitely many solution.

10) Verify the Cayley-Hamilton theorem for

i) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ii) $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$

iii) $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ iv) $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$

Unit-II: Limits and Continuity

Five marks questions:

1) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then prove that $\lim_{x \rightarrow a} (f(x) + g(x)) = l + m$

2) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then prove that $\lim_{x \rightarrow a} (f(x) - g(x)) = l - m$

3) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then prove that $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = l \cdot m$

- 4) (i) If $f(x) = \begin{cases} \frac{\log(1+x)}{x}, & \text{for } x \neq 0 \\ 1, & \text{for } x = 0 \end{cases}$ then show that $f(x)$ is continuous at $x = 0$.
- (ii) Discuss the continuity of the function at $x=0$ if $f(x) = \frac{e^{\frac{1}{x}}}{e^{\frac{1}{x}}+1}$ when $x \neq 0$; $f(0) = 0$.
- (iii) If $f(x) = \begin{cases} \frac{x^4 + x^3 + 2x^2}{\tan^{-1}x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$ discuss the continuity of $f(x)$ at $x = 0$.
- (iv) If $f(x) = \begin{cases} \frac{xe^{\frac{1}{x}}}{1+e^{\frac{1}{x}}}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$ discuss the continuity of $f(x)$ at the origin.
- (v) Examine the continuity of $f(x) = \begin{cases} \frac{e^{\frac{1}{x}} - e^{-\frac{1}{x}}}{e^{\frac{1}{x}} + e^{-\frac{1}{x}}}, & \text{for } x \neq 0 \\ 0, & \text{for } x = 0 \end{cases}$ at $x = 0$
- (vi) Discuss the continuity of $f(x) = \begin{cases} \frac{x^2 - x - 6}{x^2 - 2x - 3}, & \text{for } x \neq 3 \\ \frac{5}{4}, & \text{for } x = 3 \end{cases}$ at $x = 3$
- (vii) If $f(x) = \begin{cases} (1 + 3x)^{\frac{1}{x}}, & \text{for } x \neq 0 \\ e^3, & \text{for } x = 0 \end{cases}$ then show that $f(x)$ is continuous at $x = 0$
- 5) (i) If $f(x) = \frac{x(e^{\frac{1}{x}} - e^{-\frac{1}{x}})}{(e^{\frac{1}{x}} + e^{-\frac{1}{x}})}$ for $x \neq 0$ and $f(0) = 10$ then show that $f(x)$ is not derivable at $x=0$.
- (ii) If $f(x) = \frac{x(e^{\frac{1}{x}} - 1)}{(e^{\frac{1}{x}} + 1)}$ for $x \neq 0$ and $f(0) = 0$ then show that $f(x)$ is not differentiable at $x=0$.
- (iii) If $f(x) = \begin{cases} x \cos\left(\frac{1}{x}\right), & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$ is continuous at $x = 0$, but not differentiable at that point.
- (iv) Prove that $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$ is continuous and differentiable at $x = 0$
- 6) Prove that every continuous function $f(x)$ in $[a, b]$ is bounded in that interval.
- 7) State and prove intermediate value theorem.
- 8) Prove that a function $f(x)$ is continuous on $[a, b]$ then it attains its bounds.
- Or** Prove that a function $f(x)$ is continuous on $[a, b]$ then it attains its supremum and infimum at least once in $[a, b]$.
- 9) i) Show that $f(x) = x^2$ takes the value 5 on $[1, 3]$ using intermediate value theorem.
 ii) Show that $f(x) = x^3$ takes the value 10 on $[2, 3]$ using intermediate value theorem.
 iii) Show that $f(x) = 2x + 1$ takes the value 8 on $[2, 5]$ using intermediate value theorem.
 iv) Show that $f(x) = \sin x$ takes the value 0.5 on $[0, \frac{\pi}{2}]$ using intermediate value theorem.
 v) Show that $f(x) = e^x$ takes the value 5 on $[1, 2]$ using intermediate value theorem.
 vi) Show that $f(x) = x^2 + x$ takes the value 8 on $[1, 3]$ using intermediate value theorem.
- 10) i) Show that $x^2 - 2x - 5 = 0$ has at least one root in $(2, 3)$ using intermediate value theorem.
 ii) Show that $x^2 - 5 = 0$ has a root in $(2, 3)$ using intermediate value theorem.
 (iii) Show that $x^2 - x - 12 = 0$ has a root in $(3, 5)$ using intermediate value theorem.
- 11) Evaluate
- i) $\lim_{x \rightarrow 0} \left(\frac{\tan x - \sin x}{x^3} \right)$ iii) $\lim_{x \rightarrow 0} \left(\frac{\log \sin x}{\log \sin 2x} \right)$ iv) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}}$
 ii) $\lim_{x \rightarrow 0} \left(\frac{\log \tan x}{\log x} \right)$

$$\begin{aligned} \text{v)} \quad & \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}} \\ \text{vi)} \quad & \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} \\ \text{vii)} \quad & \lim_{x \rightarrow 0} (\sin x)^{\tan x} \\ \text{viii)} \quad & \lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \cot^2 x \right] \end{aligned}$$

$$\begin{aligned} \text{ix)} \quad & \lim_{x \rightarrow 0} (\cot x)^{\sin x} \\ \text{x)} \quad & \lim_{x \rightarrow 0} \sin x \log x \\ \text{xi)} \quad & \lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right) \\ \text{xii)} \quad & \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right) \end{aligned}$$

$$\begin{aligned} \text{xiii)} \quad & \lim_{x \rightarrow 1} x^{\frac{1}{x-1}} \\ \text{xiv)} \quad & \lim_{x \rightarrow 0} \frac{(x - \tan x)}{x^3} \\ \text{xv)} \quad & \lim_{x \rightarrow 0} \left[\frac{e^x - e^{-x} - 2x}{x^2 \sin x} \right] \end{aligned}$$

Unit-III: Polar Coordinates

Five marks questions:

- 1) With usual notations prove that $\tan \phi = r \frac{d\theta}{dr}$
- 2) Prove that the following curves cut each other orthogonally
 - i) $r = a(1 + \cos \theta)$ and $r = b(1 - \cos \theta)$
 - ii) $r = \frac{a}{(1 + \cos \theta)}$ and $r = \frac{b}{(1 - \cos \theta)}$
 - iii) $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$
 - iv) $r = a(1 + \sin \theta)$ and $r = b(1 - \sin \theta)$
 - v) $r = a\theta$ and $r = \frac{a}{\theta}$
- 3) Find the angle of intersection between the curves
 - i) $r = \sin \theta + \cos \theta$ and $r = 2 \sin \theta$
 - ii) $r = a$ and $r = a(1 + \cos \theta)$
 - iii) $r = a$ and $r = 2a \cos \theta$
 - iv) $2r = a$ and $r = a \cos \theta$
- 4) With usual notations prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$
- 5) Show that for the curve $r^2 = a^2 \sec 2\theta$, the length of the perpendicular from the pole to the tangent is $a \sqrt{\cos 2\theta}$.
- 6) Find the length of the perpendicular from the pole to the tangent for the curve $r^2 = a^2 \cos 2\theta$.
- 7) Find the pedal equation of the curve
 - i) $r = a(1 - \cos \theta)$
 - ii) $r^n = a^n \sin n\theta$
 - iii) $r = a(1 + \cos \theta)$
 - iv) $r = ae^{\theta \cot \alpha}$
- 8) For any cartesian curve $y=f(x)$, prove that $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$
- 9) For the polar curve $r = f(\theta)$, show that $\frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2}$
- 10) Find $\frac{ds}{d\theta}$ for the polar curve $r = a(1 - \cos \theta)$.
- 11) Find $\frac{ds}{dx}$ for the curve $y^2 = 4ax$.
- 12) Find $\frac{ds}{dt}$ for the parametric curves $x = a(1 + \cos t)$, $y = a(t + \sin t)$
- 13) If the curve $r = a\theta$, then show that $\frac{dr}{ds} = \frac{a}{\sqrt{r^2 + a^2}}$
 Or If the curve $r = a\theta$, then show that $\frac{ds}{dr} = \frac{\sqrt{r^2 + a^2}}{a}$

- 14) Derive the formula for the radius of curvature in the Cartesian form at any point (x,y).
- 15) Find the radius of curvature of the curve $y = \frac{a}{2} \left(e^{\frac{x}{a}} + e^{-\frac{x}{a}} \right)$ and prove that radius of curvature varies as the square of the ordinate.
- 16) Show that the radius of curvature at $\left(\frac{3a}{2}, \frac{3a}{2} \right)$ on the curve $x^3 + y^3 = 3axy$ is $\frac{3a}{8\sqrt{2}}$
- 17) Find the radius of curvature at any point (x,y) for the curves
 - i) $xy = c^2$
 - ii) $x = a \cos^3 \theta, y = a \sin^3 \theta$
 - iii) $x = a(\theta + \sin \theta), y = a(1 - \cos \theta)$
 - iv) $x = a(\cos t + t \sin t), y = a(\sin t - t \cos t)$
- 18) Derive the formula for finding the coordinates of center of curvature for the curve $y = f(x)$ at any point (x,y) on it.
- 19) Find the coordinates of the center of curvature for rectangular parabola $2xy = c^2$ at a point (c, c).
- 20) (i) Find the coordinates of the center of curvature at any point (x,y) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 (ii) Find the coordinates of the center of curvature at any point (x,y) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- 21) Find the radius and coordinates of the center of the curvature for the following curves
 - i) $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at a point $\left(\frac{a}{4}, \frac{a}{4} \right)$
 - ii) $\sqrt{x} + \sqrt{y} = 1$ at a point $\left(\frac{1}{4}, \frac{1}{4} \right)$
 - iii) $2xy + x + y = 4$ at a point (1,1)
 - iv) $xy = c^2$ at a point (c, c)
 - v) $x^3 + y^3 = 2xy$ at a point (1,1)
- 22) Find the ranges of values of x for which $y = x^4 - 6x^3 + 12x^2 + 5x + 7$ is concave upwards or concave downwards and also find points of inflexion.
- 23) Find the ranges of values of x for which $y = 3x^5 - 40x^3 + 3x - 20$ is concave upwards or downwards.
- 24) Find the points of inflexion of the curve $y = x^2 e^{-x}$.

Unit-IV: Successive Differentiation and Mean Value Theorems

Five marks questions:

- 1) State and prove Leibnitz's theorem to find the nth derivative of product of two functions.
- 2) Find the n^{th} derivative of $e^{ax} \cos(bx + c)$.
- 3) Find the n^{th} derivative of $e^{ax} \sin(bx + c)$.
- 4) If $y = x^{n-1} \log x$ then show that $y_n = \frac{(n-1)!}{x}$
- 5) If $y = a \cos(\log x) + b \sin(\log x)$ then prove that $x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$.
- 6) If $y = (\sin^{-1} x)^2$ then prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$.
- 7) (i) If $y = \cos(m \sin^{-1} x)$ then prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$.
 (ii) If $y = \sin(m \sin^{-1} x)$ then prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$.

- 8) (i) If $y = e^{m \cos^{-1} x}$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.
(ii) If $y = e^{m \sin^{-1} x}$ then show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$
(iii) If $y = e^{\tan^{-1} x}$ show that $(1 + x^2)y_{n+2} + [2(n + 1)x - 1]y_{n+1} + n(n + 1)y_n = 0$
- 9) If $x = \sin t, y = \cos pt$ then show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (p^2 - n^2)y_n = 0$
- 10) If $y = [x + \sqrt{1 + x^2}]^m$ then prove that $(1 + x^2)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$
- 11) If $y = (x^2 - 1)^n$ then prove that $(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n + 1)y_n = 0$
- 12) If I_n is the n^{th} derivative of $x^n \log x$ then prove that $I_n - nI_{n-1} = (n - 1)!$
- 13) State and prove Rolle's theorem.
- 14) State and prove Lagrange's Mean Value Theorem.
- 15) State and prove Cauchy's Mean Value Theorem.
- 16) Verify Lagrange's Mean Value Theorem for $f(x) = (x + 1)(x + 2)$ in $[1, 2]$.
- 17) Verify Cauchy's Mean Value theorem for $f(x) = \sin x, g(x) = \cos x$ in $\left[0, \frac{\pi}{2}\right]$
- 18) (i) By using Maclaurin's theorem prove that

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \text{ where } -1 < x \leq 1.$$

- (ii) By using Maclaurin's series prove that $\log(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots$
- 19) (i) By using Maclaurin's theorem prove that: $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$
(ii) By using Maclaurin's theorem prove that : $\log \sec x = \frac{x^2}{2} + \frac{x^4}{12} + \frac{x^6}{45} + \dots$
(iii) By using Maclaurin's theorem prove that $\log(1 + \sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \dots$
- 20) (i) Expand $\log(\sec x + \tan x)$ as far as term containing x^5 using Maclaurin's theorem.
(ii) Expand $\log(1 + e^x)$ as far as term containing x^4 using Maclaurin's theorem.
- 21) Expand $e^{\sin x}$ as far as term containing x^4 using Maclaurin's series.

Or By using Maclaurin's series prove that $e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots$